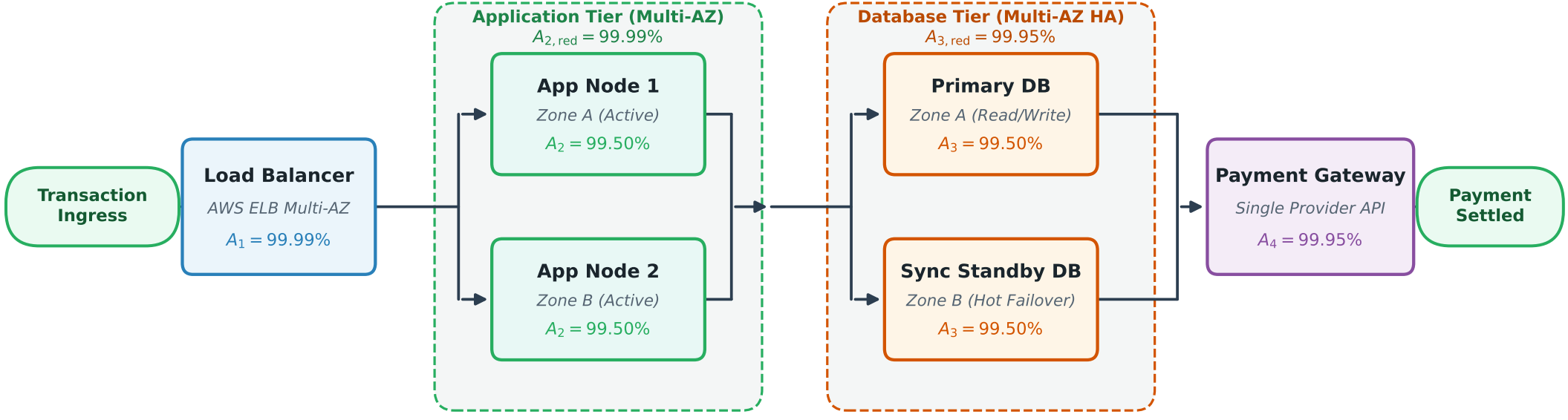
Scenario 2: Multi-AZ Redundant Architecture (Series-Parallel RBD)

$A_{red} = A_1 \cdot A_{2,red} \cdot A_{3,red} \cdot A_4 = 99.880\% \implies$ 10.51 h/year downtime (−88.6% vs. baseline)



Bottleneck Shift: With internal tiers redundant, the external Payment Gateway (PGW) becomes the dominant single point of failure (4.38 h/yr).